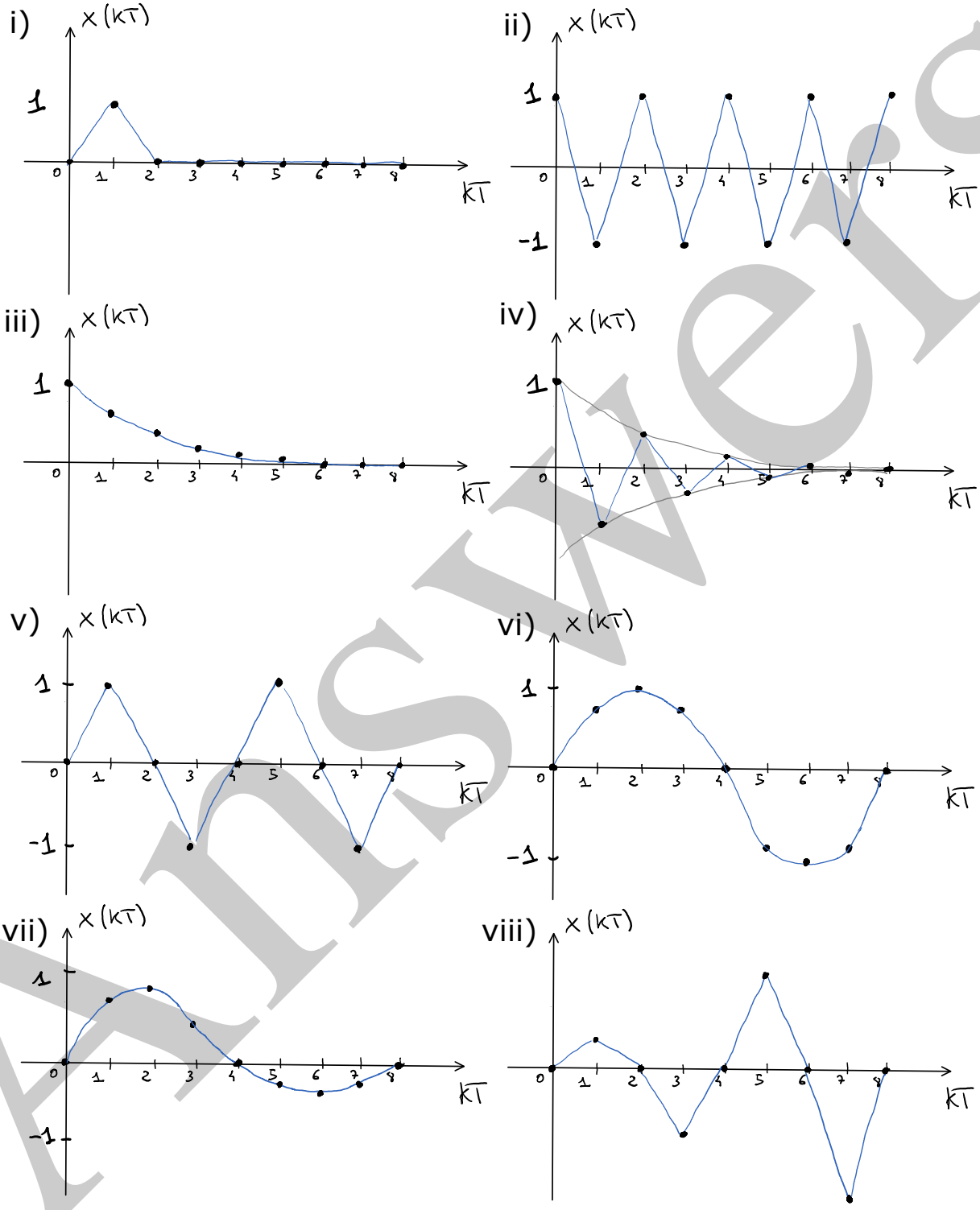
DIGITAL CONTROL SYSTEMS

1. a) The time-sequences are plotted below.



[8 marks]

b) i) The poles of the transfer function $G(s)$ are $\pm 2j$ and, consequently, the

Answers

system is marginally stable. The unit-step response of the continuous-time system is given by

$$Y(s) = \frac{s^2}{s^2+4}\frac{1}{s} = \frac{s}{s^2+4}.$$

Hence, the corresponding continuous-time signal is

$$y(t) = \cos(2t).$$

[2 marks]

ii) The discretized system is marginally stable because the discretization obtained via a zero-order hold is stability preserving. The pulse transfer function is given by

$$\begin{aligned}\frac{Y(z)}{X(z)} = Z\left[\frac{1-e^{-sT}}{s}\frac{s^2}{s^2+4}\right] &= (1-z^{-1})Z\left[\frac{s}{s^2+4}\right] \\ &= (1-z^{-1})\frac{1-z^{-1}\cos 2T}{1-2z^{-1}\cos 2T+z^{-2}}\end{aligned}$$

and the unit-step response is given by

$$Y(z) = \frac{(1-z^{-1})(1-z^{-1}\cos 2T)}{1-2z^{-1}\cos 2T+z^{-2}}\frac{1}{1-z^{-1}} = \frac{1-z^{-1}\cos 2T}{1-2z^{-1}\cos 2T+z^{-2}}.$$

[3 marks]

iii) Let $T = \dfrac{\pi}{4}$. Then

$$Y(z) = \frac{1}{1+z^{-2}} = 1 - z^{-2} + z^{-4} - z^{-6} + \ldots.$$

Hence, the first 5 samples of the sequence $y(kT)$ are

$$y(0) = 1, \quad y(T) = 0, \quad y(2T) = -1, \quad y(3T) = 0, \quad y(4T) = 1.$$

[2 marks]

iv) Let $T = \pi$. Then

$$Y(z) = \frac{1}{1+z^{-1}}$$

and $y(kT) = 1(kT)$, where $1(kT)$ is the unit step. Thus, the time-sequence is

$$y(0) = 1, \quad y(T) = 1, \quad y(2T) = 1, \quad y(3T) = 1, \quad y(4T) = 1, \quad \ldots$$

[2 marks]

v) While the sequence in part b.iv) is constant, the sequence in part b.iii) is oscillating. Hence, the continuous-time sequence is oscillating. More precisely, the discrete-time sequence is oscillating for any value of $T \neq n\pi$, with $n = 1, 2, 3, \ldots$. Thus, for the selection $T = \pi$, which does not satisfies the Nyquist criterion, the time-sequence is affected by the phenomenon of hidden oscillations. [3 marks]

Answers

2. a) i) The output is

$$y(t) = \frac{1}{T}(t+T)(1(t+T)-1(t)) + \frac{1}{T}(T-t)(1(t)-1(t-T)),$$

where $1(t)$ is the unit step. [4 marks]

ii) To find $Y(s)$ we first reorder the terms of the previous expression as

$$y(t) = \frac{1}{T}\left[(t-T)1(t-T) - 2t1(t) + (t+T)1(t+T)\right].$$

Then we apply the Laplace transform to obtain

$$Y(s) = \frac{1}{T}\left(\frac{1}{s^2}e^{-sT} - 2\frac{1}{s^2} + \frac{1}{s^2}e^{sT}\right) = \frac{e^{sT}-2+e^{-sT}}{Ts^2}.$$

Noting that $X^*(s) = 1$, it follows that

$$H_{\triangle}(s) = \frac{e^{sT}-2+e^{-sT}}{Ts^2}.$$

[4 marks]

b) We need to compute

$$Z[H_{\triangle}(s)G(s)] = Z\left[\frac{e^{sT}-2+e^{-sT}}{Ts^2}G(s)\right].$$

To this end, define $G_1(s) = \dfrac{G(s)}{Ts^2}$. Then

$$\begin{aligned} Z[H_{\triangle}(s)G(s)] &= Z\left[e^{sT}G_1(s) - 2G_1(s) + e^{-sT}G_1(s)\right] \\ &= Z\left[g_1(t+T) - 2g_1(t) + g_1(t-T)\right] \\ &= G_1(z)z - 2G_1(z) + G_1(z)z^{-1} \\ &= \left(z - 2 + z^{-1}\right)Z\left[\frac{G(s)}{Ts^2}\right] = \frac{(z-1)^2}{Tz}Z\left[\frac{G(s)}{s^2}\right]. \end{aligned}$$

Hence $f(z) = \dfrac{(z-1)^2}{Tz}$. [4 marks]

c) The frequency response of the triangle-hold is given by

$$H_{\triangle}(j\omega) = -\frac{e^{j\omega T}-2+e^{-j\omega T}}{T\omega^2}.$$

Hence,

$$\begin{aligned} H_{\triangle}(j\omega) &= \frac{1-e^{j\omega T}}{T\omega^2} + \frac{1-e^{-j\omega T}}{T\omega^2} = e^{j\frac{\omega T}{2}}\frac{e^{-j\frac{\omega T}{2}} - e^{j\frac{\omega T}{2}}}{T\omega^2} + e^{-j\frac{\omega T}{2}}\frac{e^{j\frac{\omega T}{2}} - e^{-j\frac{\omega T}{2}}}{T\omega^2} \\ &= \frac{-2je^{j\frac{\omega T}{2}}}{T\omega^2}\sin\left(\frac{T\omega}{2}\right) + \frac{2je^{-j\frac{\omega T}{2}}}{T\omega^2}\sin\left(\frac{T\omega}{2}\right) \\ &= 4\frac{\left(\sin\left(\frac{T\omega}{2}\right)\right)^2}{T\omega^2} = T\left(\frac{\sin\left(\frac{T\omega}{2}\right)}{\frac{T\omega}{2}}\right)^2. \end{aligned}$$

[4 marks]

d) The zero-order hold introduces a delay of $\dfrac{T}{2}$ seconds, the first-order hold introduces a delay of T seconds, while the triangle-hold does not introduce any delay. However, this desirable property of the triangle-hold is due to the fact that this is a non-causal first-order hold (this is clear from the impulse response in the figure). Hence, the triangle-hold is not realizable and cannot be used in practice. [4 marks]

Answers

3. a) The transfer function is

$$\begin{aligned} HP(z) &= (1-z^{-1})Z\left[\frac{1}{s^2(s+a)}\right] = \frac{(1-z^{-1})}{a^2}Z\left[\frac{-1}{s}+\frac{a}{s^2}+\frac{1}{s+a}\right] \\ &= \frac{(1-z^{-1})}{a^2}Z\left[-t1(t)+at^2 1(t)+e^{-at}1(t)\right] \\ &= \frac{(1-z^{-1})}{a^2}\left[\frac{-1}{1-z^{-1}}+\frac{aTz^{-1}}{(1-z^{-1})^2}+\frac{1}{1-e^{-aT}z^{-1}}\right]. \end{aligned}$$

After some algebra we obtain the expression

$$HP(z) = \frac{1}{a^2}\left[\frac{\left(1-e^{-aT}-aTe^{-aT}\right)+\left(aT-1+e^{-aT}\right)z}{e^{-aT}-(1+e^{-aT})z+z^2}\right].$$

[5 marks]

b) The roots of the numerator of $HP(w)$ are

$$\frac{2}{\ln(2)} > 0 \qquad \text{and} \qquad \frac{-6}{5\ln(2)-3} < 0.$$

We select $\beta = \dfrac{6}{5\ln(2)-3} \approx 12.8828$. The other possibility corresponds to a positive root and so the cancellation would make the closed-loop system unstable (in the handouts this is called "illicit cancellation"). We assume that the cancellation due to $\beta = \dfrac{6}{5\ln(2)-3}$ is exact. [4 marks]

c) The characteristic polynomial is given by

$$w^2 + \left(\frac{6}{5\ln(2)} - \alpha\frac{5\ln(2)-3}{20}\right)w + \alpha\frac{5\ln(2)-3}{10\ln(2)}.$$

Hence, the closed-loop system is asymptotically stable if

$$\frac{6}{5\ln(2)} - \alpha\frac{5\ln(2)-3}{20} > 0 \qquad \text{and} \qquad \alpha\frac{5\ln(2)-3}{10\ln(2)} > 0$$

which hold for all values of α such that

$$0 < \alpha < \frac{24}{\ln(2)(5\ln(2)-3)} \approx 74.344.$$

[4 marks]

d) The velocity constant is

$$\lim_{w\to 0} wHP(w)C(w) = \frac{3-5\ln(2)}{20}\frac{\dfrac{-2\alpha}{\ln(2)}}{\dfrac{6}{5\ln(2)}}.$$

Hence,

$$\frac{3-5\ln(2)}{20}\frac{\dfrac{-2\alpha}{\ln(2)}}{\dfrac{6}{5\ln(2)}} > 1$$

implies

$$\alpha > \frac{12}{5\ln(2)-3} \approx 25.766.$$

Answers

Thus, the range of α's for which the closed-loop system is asymptotically stable and the velocity constant is at least 1 is given by

$$25.766 \approx \frac{12}{5\ln(2)-3} < \alpha < \frac{24}{\ln(2)(5\ln(2)-3)} \approx 74.344.$$

[4 marks]

e) We select $\alpha = 2\dfrac{\beta T+2}{T} \approx 31.5365$. The discrete-time controller is

$$C(z) = [C(w)]_{w=\frac{2}{T}\left(\frac{z-1}{z+1}\right)} = \frac{\alpha T(z+1)}{(2+\beta T)z+\beta T-2} = \frac{2(z+1)}{z+\frac{\beta T-2}{\beta T+2}} \approx \frac{2(z+1)}{z+0.634}.$$

The value of $\alpha = 2\dfrac{\beta T+2}{T} \approx 31.5365$ stabilizes the closed-loop system in the w-plane and achieves a velocity constant of at least 1 in the w-plane. Owing to the correspondence between the points in the w-plane and the ones in the z-plane, it can be concluded that the discrete-time closed-loop control system is asymptotically stable. [3 marks]

Answers

4. a) The relations between the matrices (A,B,C) and (F,G,H) are

$$A = e^{FT}, \qquad B = \int_0^T e^{F\lambda} G d\lambda, \qquad C = H.$$

To compute these matrices we first compute $(sI-F)^{-1}$, namely

$$(sI-F)^{-1} = \begin{bmatrix} \frac{1}{s} & \frac{1}{s(s+2)} \\ 0 & \frac{1}{s+2} \end{bmatrix}.$$

It follows that

$$A = e^{FT} = \left(\mathscr{L}^{-1}\left[(sI-F)^{-1}\right]\right)_{t=T} = \begin{bmatrix} 1 & \frac{1}{2}\left(1-e^{-2T}\right) \\ 0 & e^{-2T} \end{bmatrix},$$

$$B = \int_0^T e^{F\lambda} G d\lambda = \begin{bmatrix} \frac{1}{2}\left(T + \frac{e^{-2T}-1}{2}\right) \\ \frac{1}{2}\left(1-e^{-2T}\right) \end{bmatrix}.$$

[4 marks]

b) To compute the transfer function we first determine the term

$$(zI-A)^{-1} = \begin{bmatrix} \frac{1}{z-1} & \frac{\frac{1}{2}\left(1-e^{-2T}\right)}{(z-1)(z-e^{-2T})} \\ 0 & \frac{1}{z-e^{-2T}} \end{bmatrix}.$$

The input-output transfer function is given by

$$\frac{Y(z)}{U(z)} = C(zI-A)^{-1}B = \frac{1}{4}\frac{(1-e^{-2T}-2Te^{-2T})+(2T-1+e^{-2T})z}{(z-1)(z-e^{-2T})}.$$

[3 marks]

c) Both the eigenvalues 1 and e^{-2T} of A appear as poles in the transfer function. Thus the system is minimal, which implies that the system is reachable and observable. [2 marks]

d) Let the estimation error be $e(k) = \tilde{x}(k+1) - x(k)$. Then

$$\begin{aligned} e(k+1) &= Ax(k) + Bu(k) - A\tilde{x}(k)) - Bu(k) - L[y(k) - C\tilde{x}(k)] \\ &= Ae(k) - LCe(k) = (A-LC)e(k). \end{aligned}$$

Thus, if the eigenvalues of the matrix $A-LC$ are inside the unit circle, then $e(k) \to 0$ as $k \to \infty$. [3 marks]

e) The desired characteristic polynomial is given by

$$(z-0.4-0.4j)(z-0.4+0.4j) = z^2 - 0.8z + 0.32.$$

The characteristic polynomial of the estimator is

$$|zI - A + LC| = z^2 + (-e^{-2T} - 1 - L_1)z + e^{-2T}L_1 + e^{-2T} - \frac{1}{2}(1-e^{-2T})$$

Answers

where $L = [L_1 \ L_2]^\top$. Thus, we have the conditions

$$\begin{aligned} -e^{-2T} - 1 - L_1 &= -0.8 \\ e^{-2T}L_1 + e^{-2T} - \frac{1}{2}(1 - e^{-2T})L_2 &= 0.32 \end{aligned}$$

which, for $T = 0.1$, have solutions $L_1 = -1.01873$ and $L_2 = -3.6999$.

[4 marks]

f) In part d) we have found that

$$e(k+1) = (A - LC)e(k).$$

Replacing $u(k) = -K\tilde{x}(k)$ into $x(k+1) = Ax(k) + Bu(k)$ we obtain

$$x(k+1) = Ax(k) - BK\tilde{x}(k) = Ax(k) - BKx(k) - BKe(k).$$

Hence, we have

$$\begin{bmatrix} e(k+1) \\ x(k+1) \end{bmatrix} = \begin{bmatrix} A - LC & 0 \\ -BK & A - BK \end{bmatrix} \begin{bmatrix} e(k) \\ x(k) \end{bmatrix}.$$

Because of the zero matrix in the top right block, the characteristic polynomial of the dynamic matrix is

$$|zI - A + LC||zI - A + BK|.$$

Thus, if K is such that the eigenvalues of $A - BK$ are inside the unit circle and L is such that the eigenvalues of $A - LC$ are inside the unit circle, then $e(k) \to 0$ and $x(k) \to 0$ as $k \to \infty$. The property that K and L can be designed independently is called *separation principle*. [4 marks]

Answers